

# Stat 17 Week 2

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## Part I: Data, Variables, and Sampling

- **Statistics and Data**

- Statistics is the science of collecting, organizing, analyzing, and interpreting data
- Data consist of observations on individuals or objects

- **Population, Sample, and Parameters**

- **Population**: the entire group of interest
- **Sample**: a subset of the population actually observed
- A **parameter** describes a population (usually unknown)
- A **statistic** summarizes a sample and estimates a parameter

- **Types of Variables**

- **Categorical/qualitative** variables place individuals into categories
  - \* Nominal: categories without order
  - \* Ordinal: categories with a natural order
  - \* Binary: two possible outcomes
- **Quantitative** variables take numerical values
  - \* Discrete: result of counting
  - \* Continuous: result of measuring

- **Sampling Methods**

- **Simple random sampling**: each individual has an equal chance of selection
- **Systematic sampling**: select every  $k$ th individual from a list
- **Stratified sampling**: divide population into groups, then randomly sample from each group
- **Cluster sampling**: randomly select groups and include all individuals within them
- **Convenience sampling**: easy to obtain but often biased

- **Bias and Ethics**

- Bias occurs when the sample is not representative of the population
- Common sources include sampling bias, response bias, and measurement bias
- Ethical data collection requires informed consent, privacy protection, and avoiding harm

## Part II: Descriptive Statistics and Data Visualization

- **Why We Summarize Data**

- Large datasets are difficult to interpret without summaries
- Numerical and graphical summaries reveal patterns and unusual features

- **Displaying Categorical Data**

- Bar charts compare frequencies or relative frequencies
- Pie charts show proportions of a whole (best with few categories)

- **Displaying Quantitative Data**

- Histograms display the distribution of numerical data
- Useful for identifying shape, center, spread, and outliers

- **Describing Distribution Shape**

- Symmetric vs. skewed distributions
- Right-skewed: long tail to the right
- Left-skewed: long tail to the left
- Outliers, gaps, and clusters may appear

- **Measures of Center**

- Mean: arithmetic average

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- Median: middle value of ordered data
- Mode: most frequent value (especially for categorical data)

- **Mean vs. Median**

- Symmetric distribution: mean  $\approx$  median
- Right-skewed: mean  $>$  median
- Left-skewed: mean  $<$  median
- Median is more resistant to outliers than the mean

- **Percentiles and Quartiles**

- The  $p$ th percentile is the value below which  $p\%$  of observations fall
- The median is the 50th percentile
- Quartiles divide ordered data into four equal parts:
  - \*  $Q_1$ : 25th percentile
  - \*  $Q_2$ : 50th percentile (median)
  - \*  $Q_3$ : 75th percentile

- **Interquartile Range (IQR)**

- The IQR measures the spread of the middle 50% of the data

$$\text{IQR} = Q_3 - Q_1$$

- The IQR is resistant to outliers

- **Outliers and the 1.5×IQR Rule**

- Lower fence:  $Q_1 - 1.5(\text{IQR})$
- Upper fence:  $Q_3 + 1.5(\text{IQR})$
- Observations outside these fences are considered outliers

- **Measures of Spread**

- Range: maximum – minimum
- Variance and standard deviation measure overall variability

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}$$

- Standard deviation has the same units as the original data

- **The Complete Description of a Distribution**

- A dataset should be described using:
  - \* Shape
  - \* Center
  - \* Spread